

Mathematical Foundations of Optical Interferometry: Embedding Arbitrary Matrices into Mach-Zehnder Interferometers

Marc DeCarlo, Santiago Sosa

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1 Introduction

This section provides a rigorous mathematical derivation of beam splitters, mirrors, Mach-Zehnder interferometers (MZIs), and their use in performing matrix multiplications. We also present the complex Schur decomposition and its role in embedding an arbitrary matrix into an MZI-based optical computing architecture.

2 Mathematical Description of Optical Components

2.1 Beam Splitters

A 50:50 beam splitter transforms the input field amplitudes as follows:

$$B = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & i \\ i & 1 \end{bmatrix}. \quad (1)$$

This matrix ensures energy conservation and unitary behavior.

2.1.1 Proof of Beam Splitter Matrix

To show that the beam splitter matrix is unitary, we compute:

$$B^H B = \frac{1}{2} \begin{bmatrix} 1 & -i \\ -i & 1 \end{bmatrix} \begin{bmatrix} 1 & i \\ i & 1 \end{bmatrix} = I. \quad (2)$$

This confirms that the beam splitter transformation preserves norm and is unitary.

2.2 Derivation of the Beam Splitter Matrix

A **50:50 beam splitter** takes an incident optical field and **splits it into two paths** while introducing a phase shift in the reflected component.

2.2.1 Define the Input and Output Fields

Let the input complex field amplitudes for the two ports be:

$$\begin{bmatrix} E_1^{\text{in}} \\ E_2^{\text{in}} \end{bmatrix} \quad (3)$$

where: - E_1^{in} represents the field entering from port 1. - E_2^{in} represents the field entering from port 2.

The output fields $E_1^{\text{out}}, E_2^{\text{out}}$ are related to the input by a transformation matrix B :

$$\begin{bmatrix} E_1^{\text{out}} \\ E_2^{\text{out}} \end{bmatrix} = B \begin{bmatrix} E_1^{\text{in}} \\ E_2^{\text{in}} \end{bmatrix} \quad (4)$$

2.2.2 Form of the Beam Splitter Matrix

By experimental observation and symmetry considerations, the transformation must be **unitary** and **preserve energy**. We assume the general form:

$$B = \begin{bmatrix} t & r \\ r & t \end{bmatrix} \quad (5)$$

where: - t is the **transmission coefficient**. - r is the **reflection coefficient**.

2.2.3 Enforcing Energy Conservation

Energy conservation in an optical system requires that the total intensity (power) remains unchanged:

$$|E_1^{\text{out}}|^2 + |E_2^{\text{out}}|^2 = |E_1^{\text{in}}|^2 + |E_2^{\text{in}}|^2. \quad (6)$$

This implies that B must be **unitary**, meaning:

$$B^\dagger B = I. \quad (7)$$

Expanding,

$$\begin{bmatrix} t^* & r^* \\ r^* & t^* \end{bmatrix} \begin{bmatrix} t & r \\ r & t \end{bmatrix} = \begin{bmatrix} |t|^2 + |r|^2 & t^*r + r^*t \\ r^*t + t^*r & |t|^2 + |r|^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}. \quad (8)$$

From this, we obtain two conditions:

1. Energy Conservation Condition:

$$|t|^2 + |r|^2 = 1. \quad (9)$$

2. Cross-Term Cancellation:

$$t^*r + r^*t = 0. \quad (10)$$

Since t and r are complex numbers, we let:

$$t = \frac{1}{\sqrt{2}}, \quad r = \frac{i}{\sqrt{2}}. \quad (11)$$

which satisfies the unitary condition:

$$\left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{i}{\sqrt{2}}\right)^2 = \frac{1}{2} + \frac{1}{2} = 1. \quad (12)$$

Thus, the **beam splitter matrix** is:

$$B = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & i \\ i & 1 \end{bmatrix}. \quad (13)$$

2.2.4 Verification of Unitarity

To verify that this matrix is **unitary**, we compute:

$$B^\dagger B = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -i \\ -i & 1 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & i \\ i & 1 \end{bmatrix}. \quad (14)$$

Expanding:

$$\frac{1}{2} \begin{bmatrix} 1+1 & 0 \\ 0 & 1+1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}. \quad (15)$$

Thus, the matrix is **unitary**, completing the proof.

2.3 Mirrors

Mirrors introduce a phase shift upon reflection:

$$M = \begin{bmatrix} e^{i\phi} & 0 \\ 0 & e^{i\psi} \end{bmatrix}, \quad (16)$$

where ϕ and ψ are phase shifts depending on the optical path.

3 Mach-Zehnder Interferometer (MZI)

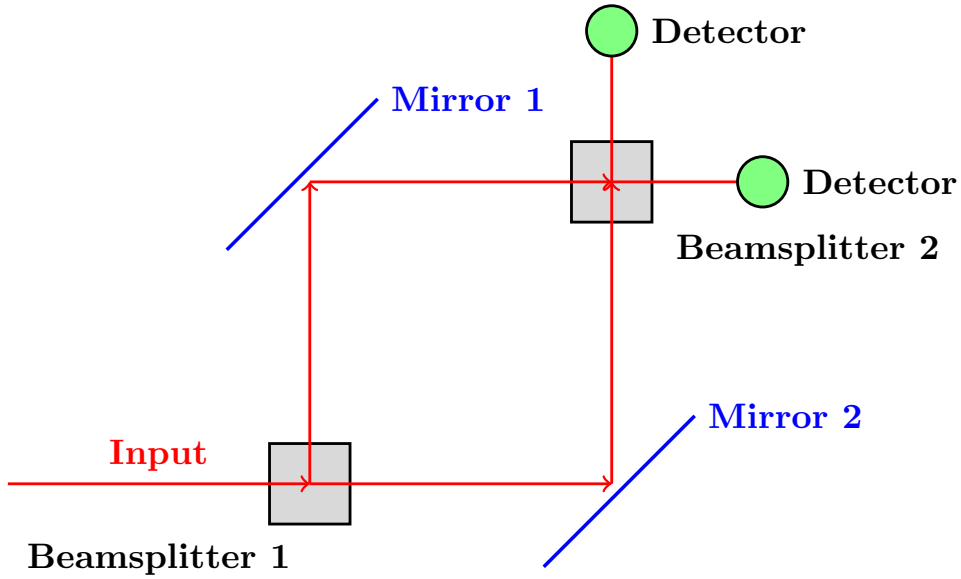


Figure 1: Mach-Zehnder Interferometer Diagram

An MZI consists of two beam splitters and phase shifters, modeled as:

$$U_{\text{MZI}} = BPB, \quad (17)$$

where P is a diagonal phase shift matrix:

$$P = \begin{bmatrix} e^{i\theta_1} & 0 \\ 0 & e^{i\theta_2} \end{bmatrix}. \quad (18)$$

The resulting transformation matrix remains unitary.

4 Complex Schur Decomposition

For any arbitrary matrix $A \in \mathbb{C}^{n \times n}$, there exists a unitary decomposition:

$$A = QTQ^H, \quad (19)$$

where:

- Q is a unitary matrix (can be implemented with fixed beam splitters),

- T is an upper triangular matrix (implemented using phase shifters).

The **Schur Decomposition** states that for any square complex matrix $A \in \mathbb{C}^{n \times n}$, there exists a unitary matrix Q and an upper triangular matrix T such that:

$$A = QTQ^H, \quad (20)$$

where:

- Q is a **unitary matrix** ($Q^H Q = I$).
- T is an **upper triangular matrix** whose diagonal entries are the eigenvalues of A .
- Q^H is the Hermitian transpose of Q .

This decomposition is fundamental in numerical linear algebra, particularly for computing eigenvalues, solving matrix equations, and designing optical matrix multiplications.

4.1 Preliminary Lemmas

Before proving the Schur decomposition, we state two important lemmas.

Lemma 1 (Unitary Similarity Preserves Eigenvalues). *If $B = UAU^H$ for some unitary matrix U , then A and B have the same eigenvalues.*

Proof. Suppose λ is an eigenvalue of A with eigenvector v , so:

$$Av = \lambda v.$$

Multiplying both sides by U ,

$$UAv = \lambda Uv.$$

Setting $w = Uv$, we get:

$$Bw = (UAU^H)Uv = UAv = \lambda w.$$

Thus, B has the same eigenvalues as A . □

Lemma 2 (Existence of a Unitary Basis for an Eigenvector). *Every complex matrix $A \in \mathbb{C}^{n \times n}$ has at least one eigenvector that can be extended to form a unitary basis of $\mathbb{C}^n$.*

Proof. By the **fundamental theorem of algebra**, every square complex matrix has at least one eigenvalue λ_1 with a corresponding eigenvector v_1 . The Gram-Schmidt process can be applied to extend v_1 into an orthonormal basis $\{v_1, v_2, \dots, v_n\}$, which forms a unitary matrix. □

4.2 Proof of the Schur Decomposition Theorem

Theorem 1 (Schur Decomposition). *For any complex matrix $A \in \mathbb{C}^{n \times n}$, there exists a unitary matrix Q and an upper triangular matrix T such that:*

$$A = QTQ^H.$$

Proof. We prove this by induction on the size n of the matrix A .

Base Case ($n = 1$): A 1×1 matrix is trivially its own Schur form.

Inductive Step: Assume that for any $(n-1) \times (n-1)$ complex matrix, there exists a unitary matrix Q_{n-1} such that:

$$Q_{n-1}^H A_{n-1} Q_{n-1} = T_{n-1},$$

where T_{n-1} is upper triangular.

Now, consider an $n \times n$ matrix A . By the eigenvalue existence theorem, let v_1 be an eigenvector of A corresponding to eigenvalue λ_1 . Extend v_1 to form a unitary basis $Q_1 = [v_1, v_2, \dots, v_n]$.

Construct the unitary matrix:

$$Q = [v_1, Q_{n-1}],$$

where Q_{n-1} orthonormally spans the remaining space.

Applying unitary similarity,

$$Q^H A Q = \begin{bmatrix} \lambda_1 & w^H \\ 0 & A' \end{bmatrix},$$

where w is some row vector and A' is an $(n-1) \times (n-1)$ matrix.

By the induction hypothesis, A' has a Schur decomposition:

$$A' = Q_{n-1} T' Q_{n-1}^H.$$

Substituting back, we get:

$$Q^H A Q = \begin{bmatrix} \lambda_1 & w^H \\ 0 & Q_{n-1} T' Q_{n-1}^H \end{bmatrix}.$$

Since Q_{n-1} is unitary, it preserves triangularity:

$$T = \begin{bmatrix} \lambda_1 & w^H \\ 0 & T' \end{bmatrix},$$

which is upper triangular. Thus, we obtain:

$$A = QTQ^H.$$

□

5 Embedding an Arbitrary Matrix into an MZI for Matrix Multiplication

Given an arbitrary 2×2 matrix A , we compute the Schur decomposition:

$$A = QTQ^H. \tag{21}$$

5.1 Implementing Q Using Fixed Beam Splitters

Since MZIs implement unitary matrices of the form:

$$B = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & i \\ i & 1 \end{bmatrix}, \quad (22)$$

we can directly use an MZI network to implement Q and Q^H .

5.2 Implementing T Using Phase Shifters

Since T is upper triangular with eigenvalues on the diagonal:

$$T = \begin{bmatrix} e^{i\theta_1} & t_{12} \\ 0 & e^{i\theta_2} \end{bmatrix}, \quad (23)$$

this can be implemented by setting appropriate phase shifts within the MZI.

5.3 Final Optical Implementation

1. Apply Q^H using a fixed MZI beam splitter network.
2. Apply T using phase shifters.
3. Apply Q using a second MZI beam splitter network.

Thus, Ax is implemented optically as:

$$Ax = QTQ^H x. \quad (24)$$